

# Joseph Milliken

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## EDUCATION

**Bachelor of Science in Computer Science and Engineering**, *University Of Colorado Denver* 08/2022 – 05/2026 | Denver, CO  
GPA: 3.94/4.0

## PROFESSIONAL EXPERIENCE

**Teaching Assistant | Algorithms & Principles of Programming Languages** 08/2024 – 05/2026 | Denver, CO  
*University of Colorado Denver*

- Assisted in teaching undergraduate courses in algorithms and programming languages, reinforcing concepts such as **computational complexity**, dynamic programming, greedy algorithms, **language semantics**, and type systems.
- Led regular office hours and provided individualized academic support, helping students improve comprehension, and problem solving skills.

## PROJECTS

**Clara, Senior Capstone Project**, *MERN, Python, AWS S3*

- Engineered an end-to-end **AI platform** for whole-slide image (WSI) biopsy classification, leveraging **PyTorch**, **FastAPI**, **Docker**, and **Amazon S3** to support pathology workflows.
- Applied **Agile** software development practices within a four person team over two semesters to deliver a clinical decision platform that reduced pathology second-opinion turnaround times from up to **two weeks** to approximately **five seconds**.
- Earned **1st Place** among Computer Science teams at the College of Engineering Senior Capstone Expo for developing an AI-powered clinical decision-support platform with healthcare impact.

**File Sharer**, *C, Socket Programming, Avalonia UI*

- Developed a **peer-to-peer** file sharing system in C using **UDP sockets** and **multicast networking**, enabling distributed file discovery and direct client-to-client file transfers.
- Designed a chunk-based file distribution protocol that partitions files into fixed-size segments, generates **SHA-256 hashes** for integrity verification, and reconstructs downloaded files with validation against original file hashes.

**taskctl | Golang, JSON**

- Developed a lightweight, **cross-platform** command-line task manager.
- Implemented core task operations: add, update, delete, mark complete/incomplete, and clear all tasks.
- Enhanced usability with features like priority and due dates, sorting by status/priority/date.
- Designed a polished interactive UI with promptui, supporting intuitive keyboard navigation.

## SKILLS

**Languages:** Python, C/C++, Golang, C#, SQL, Javascript/Typescript, CSS, HTML, Bash

**Developer Tools:** Git/Github, Copilot, Render, MongoDB, Visual Studio/VS Code, Arduino IDE, phpMyAdmin, Postman, ChatGPT

**Frameworks and Libraries:** Express.js, React.js, pandas, numpy, matplotlib

**Platforms:** Docker, macOS, Windows, AWS (S3), PostgreSQL

## CERTIFICATES

**NVIDIA:** Applications of AI for Anomaly Detection

**Docker:** Foundations Professional Certification

**Atlassian:** Agile Project Management Professional Certificate

**CU Denver:** Cybersecurity and Secure Computing

## AWARDS

**Dean's List**

Fall 22 | Spring 23 | Fall 23 | Spring 24 | Fall 24 | Spring 25 | Fall 25 | Spring 26

## LANGUAGES

**Spanish** — Native/Bilingual